Music Generation

Choose a representation for pitch, timing, simultaneity, and instruments before selecting a model. Symbolic music can be generated sequentially or as a multitrack grid.

Original study notes based on David Foster, Second edition, 2023. Book pp. 297–330. Source APIs, services, experiments, and model comparisons reflect the book's publication period. These are explanatory summaries, not verbatim reproductions or a claim that every source detail is retained.

MIDI score → Parse notes and durations → Token or piano-roll representation → Train → Decode and listen

Detailed Notes

Music as structured data and Bach cello suites

Source: book pp. 297–300 / PDF p. 325

Music adds challenges beyond a text stream: pitch, duration, silence, timing, harmony, voices, and long-range structure. MIDI represents symbolic events rather than the audio waveform, enabling editing and compact analysis.

The monophonic Bach cello project makes sequence modeling manageable because one note/rest event can be represented at a time. Parse scores with a suitable library such as music21, retain musical boundaries, and check that duration/pitch extraction preserves the intended score.

Parsing MIDI, tokenization, and training sets

Source: book pp. 300–304 / PDF p. 328

Convert notes, rests, and durations into explicit vocabularies, then create shifted training windows. Treat rests and note lengths as meaningful events rather than stripping them as noise.

Split by musical piece where possible to avoid nearly identical passages in both training and evaluation. Quantizing durations simplifies modeling but loses expressive timing. Record how ties, chords, and unusual duration values are handled. The representation determines which outputs are even possible, so preprocessing is part of the musical model.

Sine positional encoding and multiple streams

Source: book pp. 305–308 / PDF p. 333

The transformer needs position information to distinguish earlier and later events. Sinusoidal encodings give positions predictable feature patterns; pitch and duration embeddings can be combined while preserving their different roles.

A multi-input, multi-output architecture predicts note and duration distributions, typically with separate output heads and combined losses. Ensure target alignment: both heads must correspond to the same next musical event. Loss weighting can favor pitch while neglecting rhythm, so inspect each component and the decoded result.

Analysis of the music-generating transformer

Source: book pp. 309–312 / PDF p. 337

Sample pitch and duration under controlled temperatures, reconstruct a playable score, and listen. Visual note plots, key/scale consistency, duration distributions, repetition, and structural development complement likelihood.

Locally plausible notes do not prove good composition. Compare different seeds and lengths, inspect copied passages, and assess whether sampling independently from multiple heads creates incompatible event combinations. Human listening is essential but should be supported by repeatable sample sets rather than a single pleasing excerpt.

Polyphonic grid versus event tokenization

Source: book pp. 313–316 / PDF p. 341

Grid tokenization serializes a piano-roll-like representation of multiple voices at fixed timesteps. It makes simultaneity explicit but expands sequence length and quantizes timing.

Event tokenization uses instructions such as note-on, note-off, time shifts, and instrument/velocity changes. It can be more compact for sparse activity but requires valid ordering and event bookkeeping. Chord structure and long-range dependencies can be difficult in either form. Choose based on temporal precision, polyphony, sequence budget, and decodability.

MuseGAN data and generator

Source: book pp. 317–325 / PDF p. 345

MuseGAN treats multitrack symbolic music as a spatial tensor: bars, timesteps, pitch, and tracks/channels. The Bach chorale project demonstrates synchronized polyphonic generation rather than token-by-token prediction.

Its generator organizes shared and track-specific latent factors for chord, style, melody, and groove. Temporal components create related bar-level codes, while convolutional generators produce synchronized track grids. Shared structure is intended to coordinate instruments; private codes allow distinct voices. Match tensor dimensions carefully when converting between score and piano roll.

MuseGAN critic, analysis, and limitations

Source: book pp. 326–330 / PDF p. 354

A convolutional critic evaluates generated multitrack scores against real examples, with stable adversarial objectives such as WGAN-GP. Training needs an appropriate conversion of continuous outputs into note activity for inspection and playback.

Vary one latent factor at a time to see whether desired control exists. Assess harmony, synchronization, empty bars, pitch ranges, repetition, and diversity, not only critic loss. The example is an investigation of symbolic structure; it does not automatically synthesize realistic instrument audio.

Revision Check

1. How does MIDI differ from audio?
2. What information is lost by a fixed time grid?
3. How do separate note/duration heads stay aligned?
4. Why does polyphony complicate tokenization?
5. Which MuseGAN latents are shared across tracks?

Music quality is inseparable from representation. A model cannot recover musical distinctions that preprocessing has discarded.

Source links refer to the supplied PDF. Every contents-page topic is mapped in coverage.md. Consult the book for complete implementations, figures, exercises, citations, and experiment evidence.